

Rock Creek Watershed Digital Twin

Model description, data and validation

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Abstract

The Rock Creek digital twin forecasts streamflow, stage, soil moisture and groundwater across the Rock Creek watershed (Maryland and the District of Columbia) for the next seven days, every six hours. It runs a physically based, semi-distributed model built in the open-source OpenHydroQual (OHQ) framework (Massoudieh et al., 2023) with 54 sub-catchments, a channel network with surveyed gage sections and two flood-control reservoirs. Each sub-catchment represents its impervious and pervious land, three soil layers, a regional aquifer and a near-stream aquifer. Every input — terrain, land cover, imperviousness, soils, channel geometry, reservoir geometry, rainfall and evapotranspiration — comes from public data sets, processed by scripts that are part of the project. Twelve watershed-wide parameters were calibrated against two years of discharge and stage at the two main-stem USGS gages, and the model was then tested, without change, on six further years (WY2020–2025) at those gages and at a tributary gage that was not used in calibration. Over the same years it reproduces the gages better than NOAA’s National Water Model, most of all at low flows and on the small tributary. This document describes the model, its equations, its data and the validation results.

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1 What the twin does

Every six hours the twin

1. updates its rainfall with NOAA’s Multi-Radar/Multi-Sensor (MRMS) radar–gauge precipitation analysis (hourly, 1 km, about one hour behind real time) and its reference evapotranspiration with gridMET, followed by the most recent days of Open-Meteo;
2. advances the model from its last saved state to the present (the *analysis*);
3. runs the model seven days ahead from that state with the Open-Meteo hourly precipitation and reference-evapotranspiration forecasts (the *forecast*);
4. publishes the results: maps of each model element, time series of every variable in every sub-catchment, and model versus USGS real-time flow and stage at the gages.

The model state is carried from cycle to cycle, so the twin keeps a continuous memory of soil moisture and groundwater. It does not yet assimilate observations: the model is not adjusted to match the gages, which makes the comparison with the gages shown in the viewer an ongoing, independent check.

2 The watershed

Rock Creek drains 197.8 km^2 of the Maryland Piedmont and the District of Columbia to the Potomac River (Fig. 1). The upper basin (Montgomery County) is suburban with two flood-control and recreation reservoirs, Lake Needwood and Lake Frank; the lower basin runs through Rock Creek Park in Washington, DC. The basin is about 30 % impervious overall. Part of the District drains to the DC Water combined sewer system, whose runoff goes to the Blue Plains treatment plant rather than to the creek; this area (12.2 km^2 , mostly in the three lowest sub-catchments, Fig. 18) is excluded, leaving 185.7 km^2 that contributes to the stream. Combined-sewer overflows are not represented.

Rock Creek watershed (197.8 km²) — gages, dams and stream network

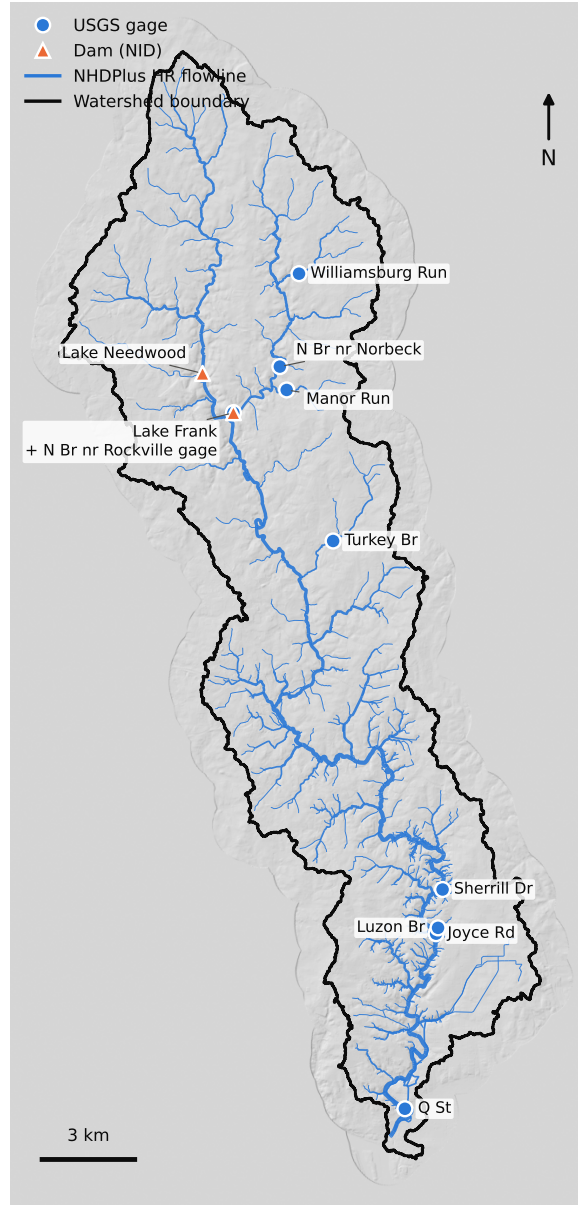


Figure 1: Rock Creek watershed with the NHDPlus High Resolution stream network, the USGS gages and the dams (National Inventory of Dams).

3 Model structure

3.1 Sub-catchments and channel network

The watershed is delineated from the USGS 3DEP 10 m elevation model, with the NHDPlus High Resolution flowlines burned in (Fig. 2), into 54 sub-catchments whose outlets are placed at the gages, the dams and the main confluences (Fig. 3). Each sub-catchment drains to one channel segment; the segments form the stream network down to the Potomac (Fig. 4). Bed elevations

follow the burned main-stem profile (Fig. 5).

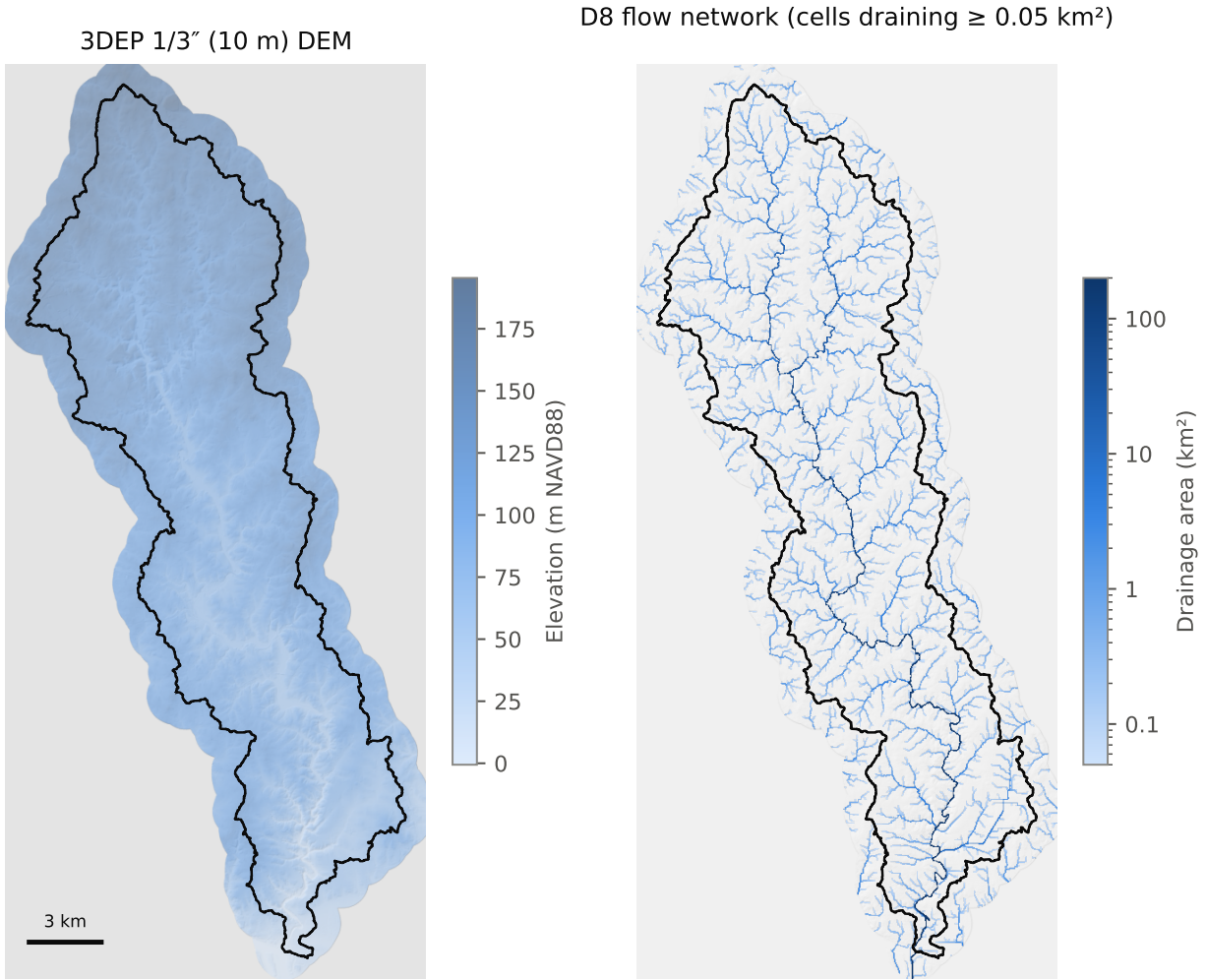


Figure 2: Left: 3DEP 10 m elevation model. Right: D8 flow network after burning in the NHD HR flowlines (cells draining at least 0.05 km^2).

54 sub-catchments and model reaches

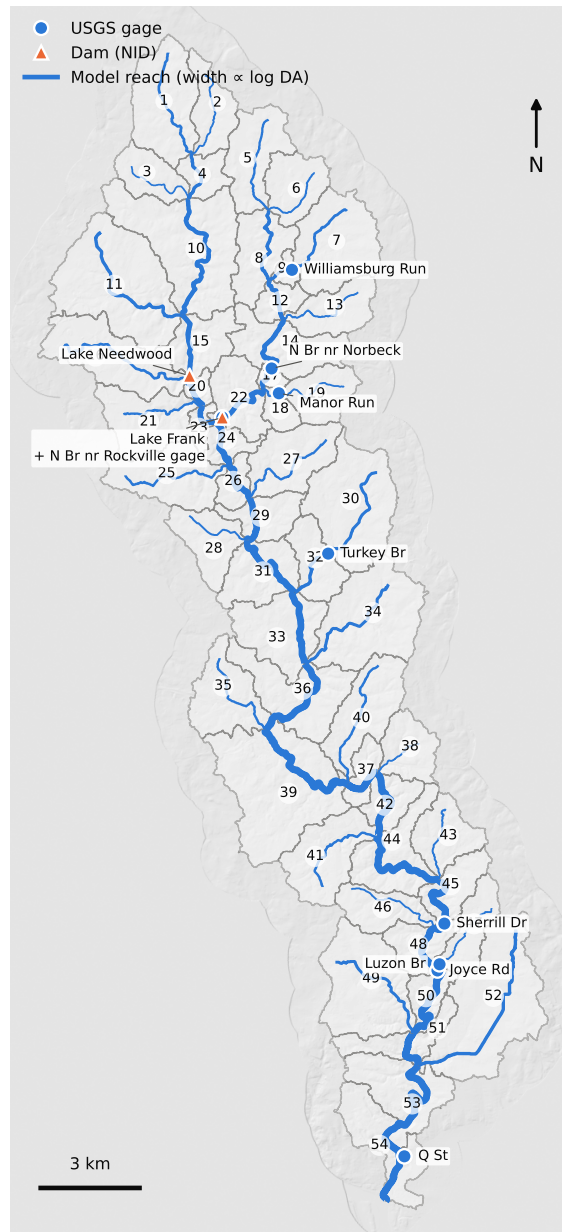


Figure 3: The 54 sub-catchments with their channel segments (line width proportional to the logarithm of drainage area) and control points.

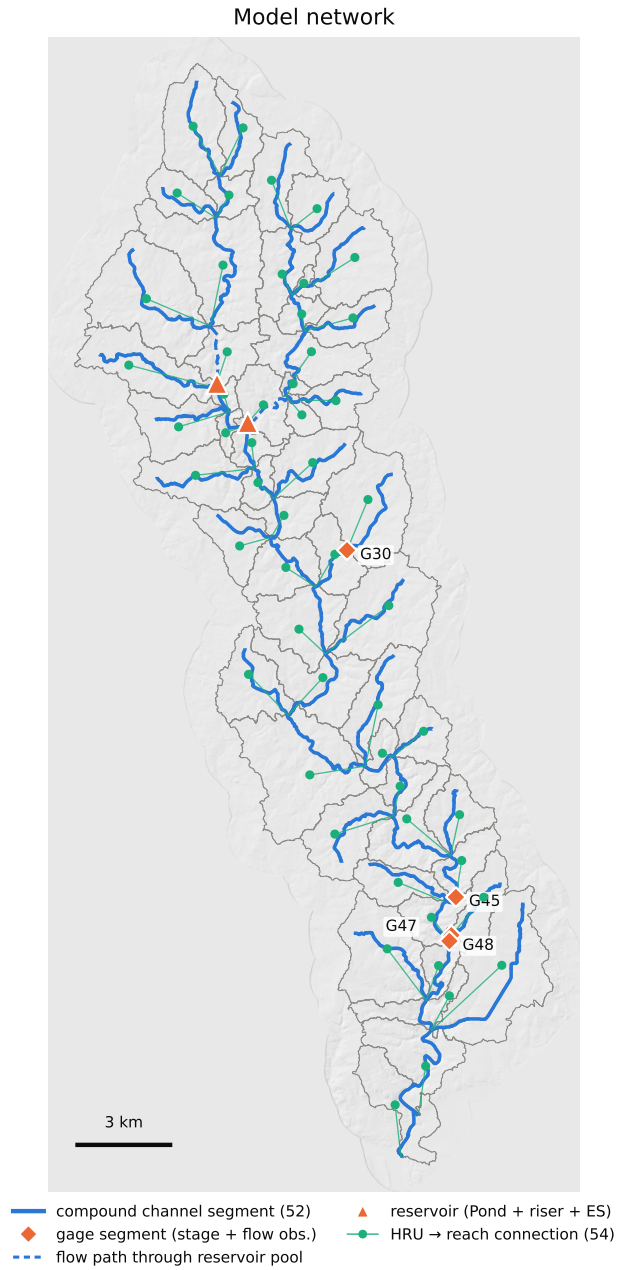


Figure 4: Model network: channel segments, gage segments, reservoirs and sub-catchment connections.

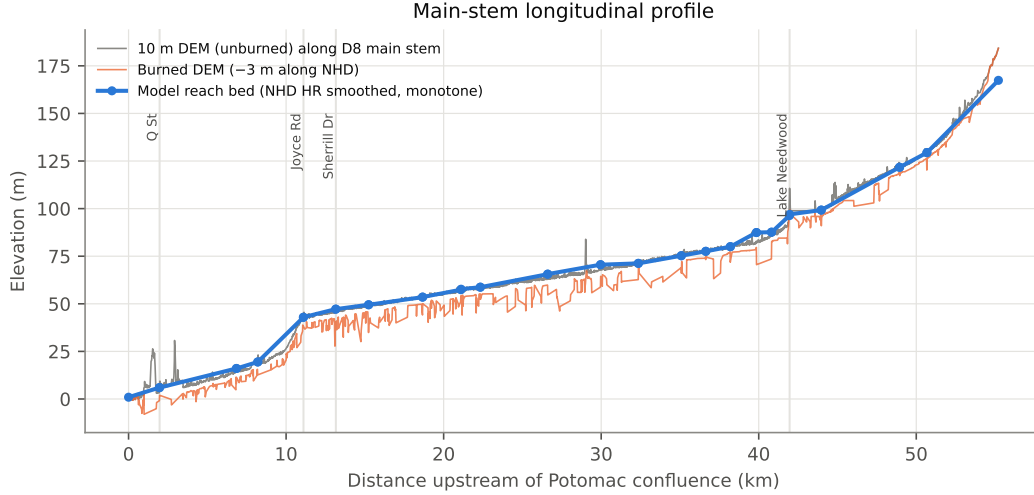


Figure 5: Main-stem longitudinal profile: elevation model along the main stem before and after burning, and the model bed elevations.

3.2 The hydrologic response unit

Each sub-catchment is one *Riparian Urban HRU*, a composite element of OHQ (Fig. 6):

- an **impervious surface** — the *directly connected* impervious area (roofs, roads and parking areas that drain to a storm sewer or the stream) — that drains through an internal impervious reach (gutters and storm pipes) to the stream;
- a **pervious surface** — lawns, forest and the impervious surfaces that drain onto them — that drains through an internal pervious reach (swales and small channels);
- three **soil layers** below the pervious surface, 0.5, 1 and 2 m thick, with their own hydraulic properties;
- a **regional aquifer** below the whole unit, exchanging groundwater with the neighbouring units;
- a **near-stream aquifer**, a strip along the stream that receives water from the regional aquifer and the soil above it and exchanges water with the stream in both directions.

The near-stream aquifer gives the unit a fast and a slow groundwater store in series: it fills from the stream and from recharge during storms and drains over days to weeks, reproducing the recessions observed after each event, while the regional aquifer supplies the slow seasonal baseflow.

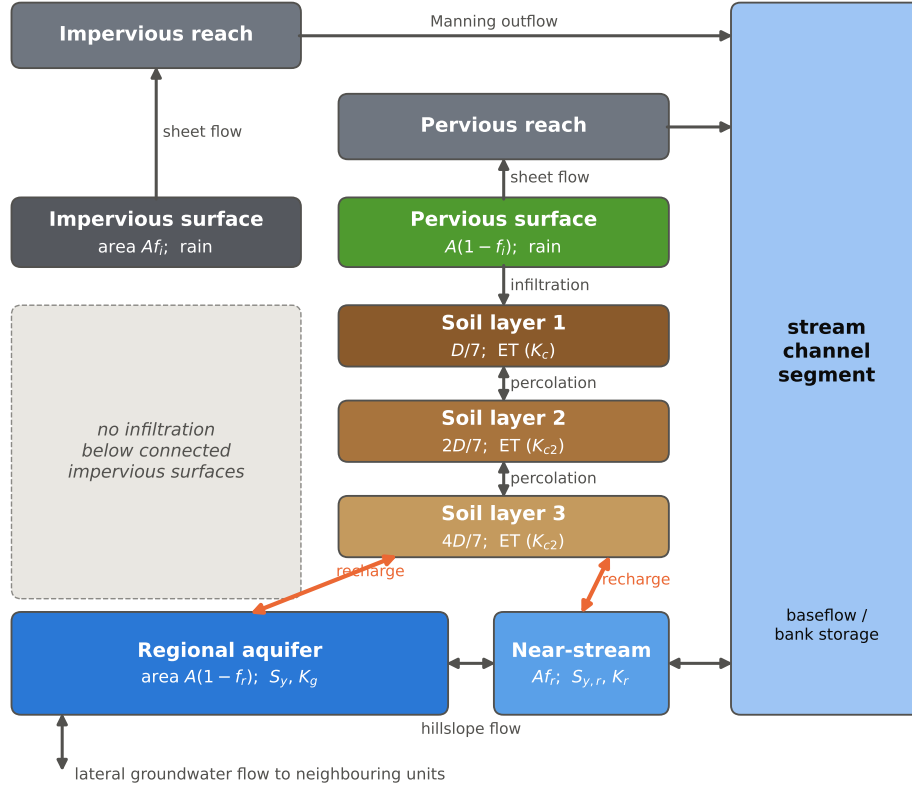


Figure 6: Structure of one sub-catchment (Riparian Urban HRU): storages (boxes) and fluxes (arrows; double arrows can flow both ways). A : area; f_i : connected impervious fraction; f_r : near-stream fraction; $D = 3.5$ m: soil depth.

3.3 Channels and gage sections

Each channel segment has a *compound* cross-section: a power-law main channel, $A = aY^m$, up to the bank depth Y_b , and above it a floodplain of width B_f and side slope z_f (Fig. 7). Main-channel coefficients come from the Maryland Piedmont bank-full relations (McCandless and Everett, 2002), floodplain widths and slopes from the 1 m lidar (Fig. 8). At each of the three USGS gages a short *gage segment* has a section fitted to the continuous record of simultaneous discharge and gage height (Fig. 9); its depth above the zero-flow control is directly comparable to the USGS stage.

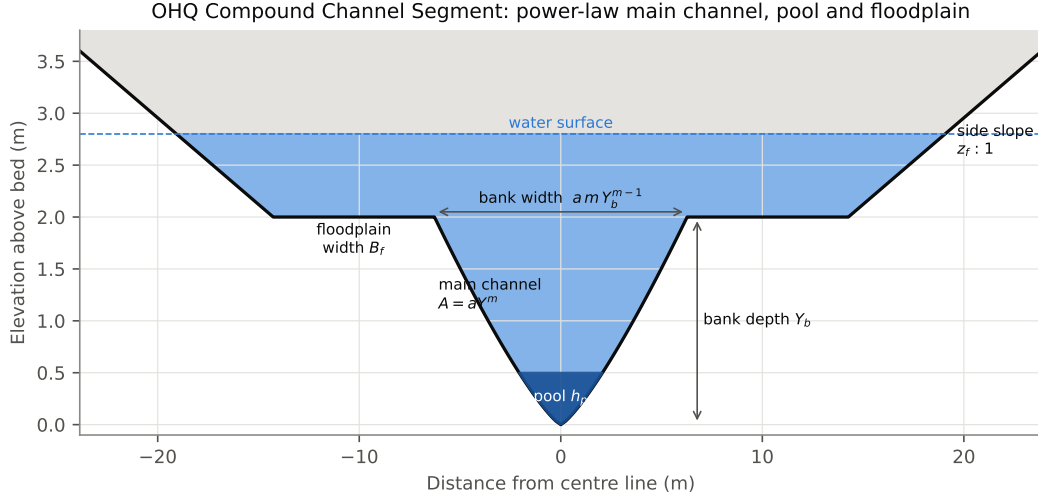


Figure 7: Compound channel cross-section.

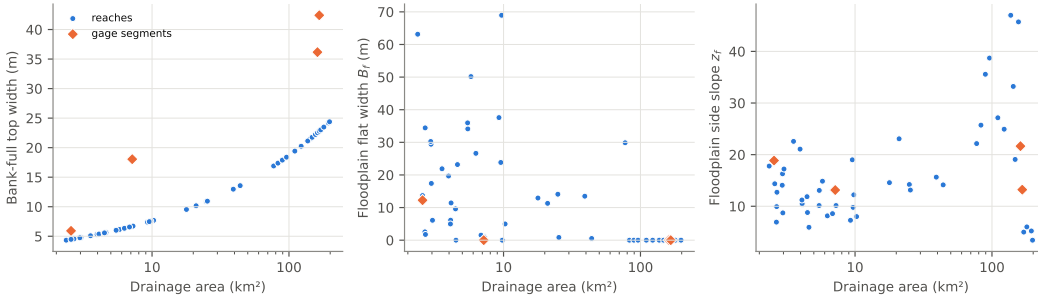


Figure 8: Bank-full width, floodplain width and floodplain side slope versus drainage area.

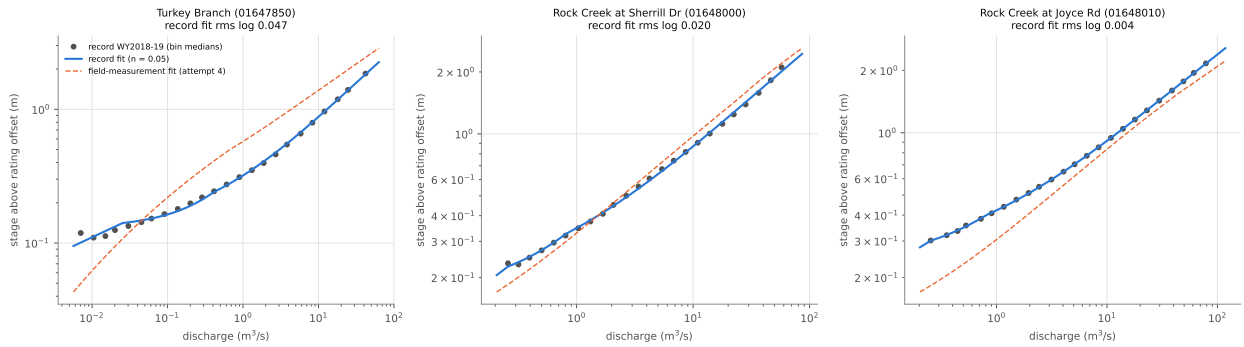


Figure 9: Gauge sections fitted to the continuous discharge–stage record.

3.4 Reservoirs

Lake Needwood and Lake Frank are storage elements with stage–storage curves from the 1 m lidar (Figs. 10 and 11), a principal (riser and conduit) spillway at the normal pool and an emergency spillway at the lowest overflow path.

1 m lidar (MD Central 2024) — Lake Needwood and Lake Frank

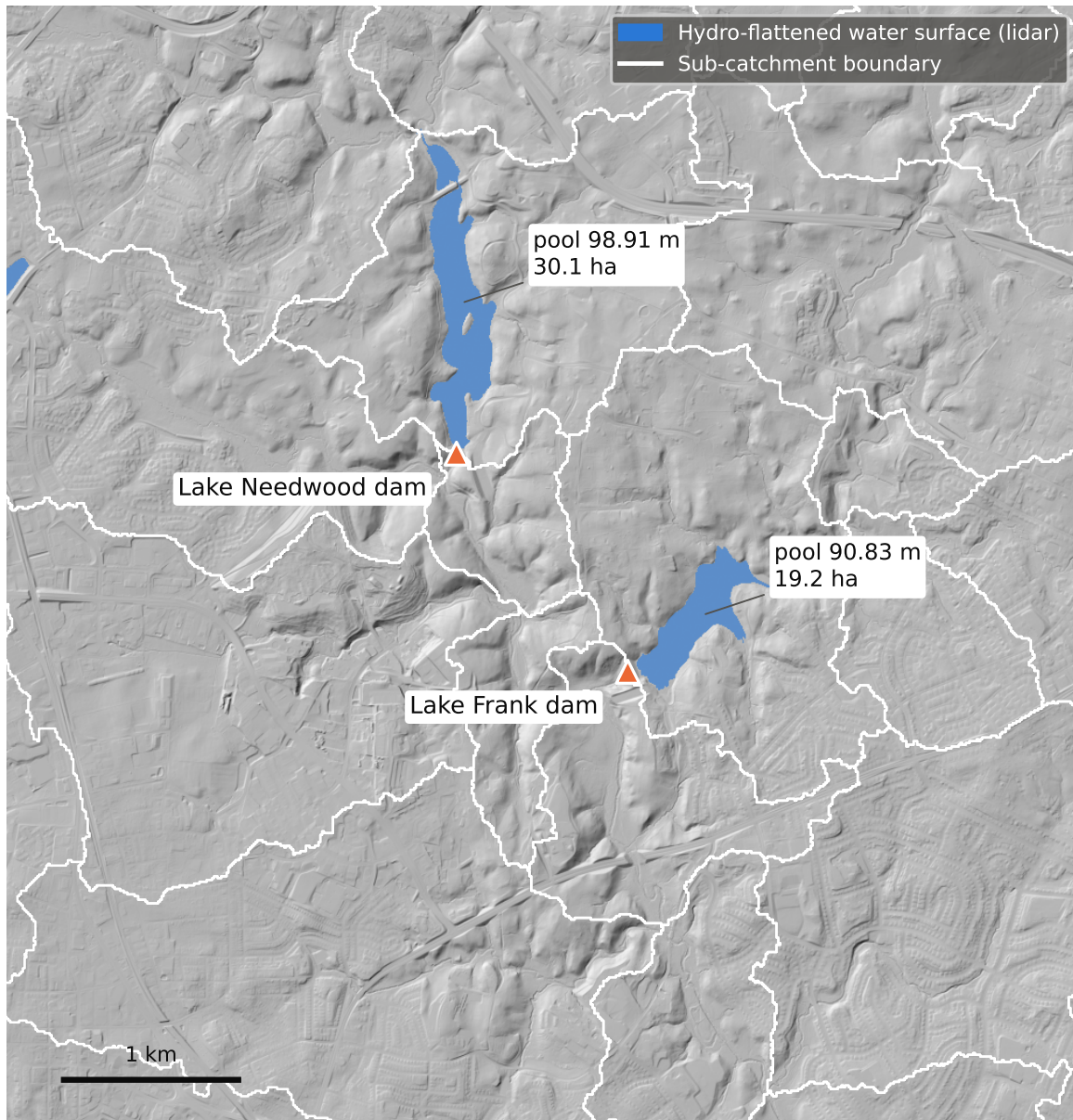


Figure 10: Lake Needwood and Lake Frank in the 1 m lidar.

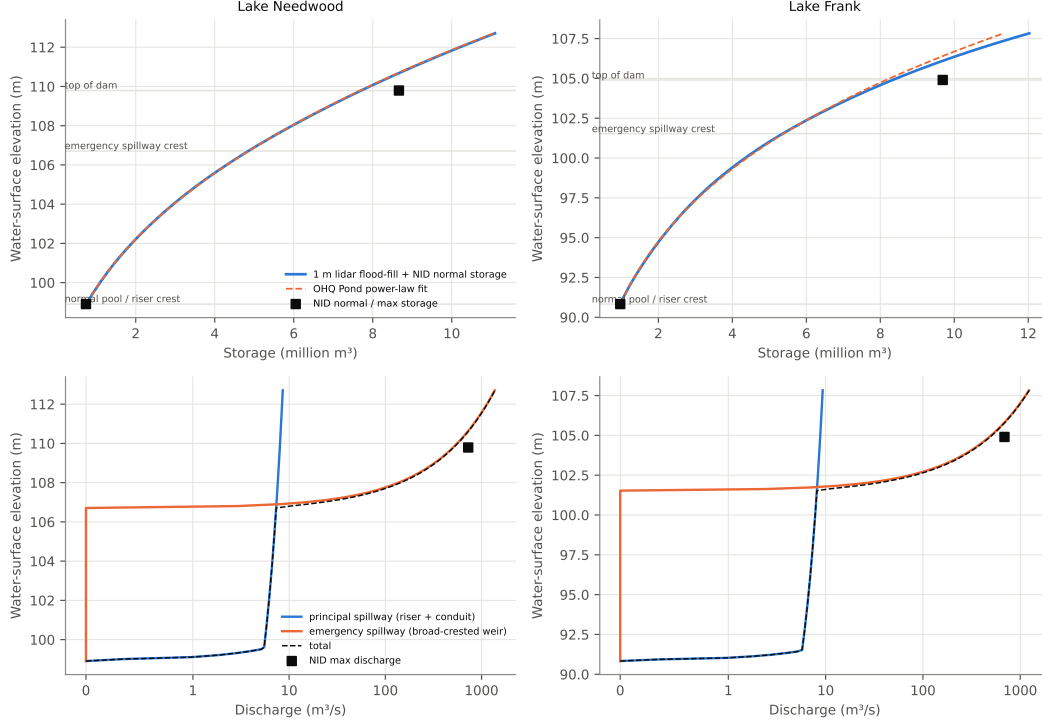


Figure 11: Stage-storage and stage-discharge of the two reservoirs.

4 Governing equations

OpenHydroQual represents a system as a network of storage elements (blocks) connected by flux elements (links), each defined by user-editable expressions, and solves the resulting equations together (Massoudieh et al., 2023; the equations of the standard components are given in Massoudieh and Shakouri, 2025). It solves a water balance for every storage element (surface, reach, soil layer, aquifer, channel segment, reservoir),

$$\frac{dV_j}{dt} = \sum_{k \in \text{in}(j)} Q_k - \sum_{k \in \text{out}(j)} Q_k + P_j - E_j, \quad (1)$$

where V_j is the stored volume, Q_k the flows of the connections (links) that enter and leave it, P_j rainfall and E_j evapotranspiration. The flows depend on the heads of the two elements they connect, as follows. Lengths are in metres and time in days; the Manning-type relations below give m^3/s and are multiplied by 86 400.

Surface runoff. Water ponded on a surface above its depression storage d_s runs off to the internal reach as sheet flow,

$$Q_s = \frac{W}{n_s} S^{1/2} (d - d_s)_+^{5/3}, \quad (2)$$

with overland-flow width W , slope S and Manning coefficient n_s ($(x)_+ = \max(x, 0)$).

Internal reaches and channels. Flow between a reach and the next segment, and between channel segments, follows Manning’s equation driven by the difference in water-surface elevation,

$$Q = \frac{1}{\bar{n}} A_u R^{2/3} \left(\frac{h_1 - h_2}{\bar{L}} \right)^{1/2}, \quad (3)$$

with the flow area A_u of the upstream end, hydraulic radius R , mean length $\bar{L}$ and mean roughness $\bar{n}$; backwater and flow reversal are therefore represented.

Soil water. Each soil layer follows the van Genuchten–Mualem relations,

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, \quad \psi = -\frac{1}{\alpha} \left(S_e^{-1/m} - 1 \right)^{1/n}, \quad k_r = \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2, \quad m = 1 - \frac{1}{n}, \quad (4)$$

with total head $h = z + \psi$. *Infiltration* into the top layer is Darcy flow over half its thickness, $Q = A K_{s,1} (h_s - h_1)/(d_1/2)$, and *percolation* between layers is $Q = A K_s k_r(S_e) (h_i - h_j)/\bar{d}$, upward when the lower layer has the higher head (capillary rise).

Evapotranspiration. Each soil layer loses

$$E = K_c E_0(t) A \frac{\theta}{\theta_s}, \quad (5)$$

where E_0 is the reference (grass) evapotranspiration and K_c a crop coefficient, one for the top layer and one for the two deeper layers.

Recharge. The bottom soil layer drains to each aquifer through its unsaturated conductivity, and receives water from it at the saturated conductivity when the water table is higher,

$$Q_r = A_r K_{s,3} \frac{k_r(S_{e,3}) (h_3 - h_g)_+ - (h_g - h_3)_+}{L}, \quad (6)$$

over the share A_r of the pervious area above that aquifer.

Aquifers. The water table of each aquifer cell follows from its storage and specific yield S_y , $h_g = z_b + V/(S_y A)$. Groundwater moves between neighbouring units, and from the regional to the near-stream aquifer, by Darcy’s law with the saturated thickness b as transmissive thickness,

$$Q = \bar{K} \bar{b} W \frac{h_1 - h_2}{L}, \quad (7)$$

with interface width W (the shared boundary between units; twice the stream length within a unit for the regional–near-stream exchange) and length L (centroid distance; the hillslope length for the regional–near-stream exchange). Lateral outflow tapers to zero as a cell approaches the depth of its stream bed, so that one unit cannot be drained by its neighbours.

Baseflow and bank storage. The near-stream aquifer exchanges water with the channel,

$$Q_b = K_r A_b \frac{h_g - h_c}{L_b}, \quad (8)$$

with bank area A_b and path length L_b ; the flow reverses when the stream is above the water table.

Reservoirs. The principal spillway discharges the lesser of the riser weir and the conduit capacity, $Q = \min(C_w L H^{3/2}, \frac{\pi}{4} D^2 \sqrt{2g \Delta h / (1 + K)})$; the emergency spillway is a broad-crested weir.

Numerical solution. The equations of all 547 storages are solved together with an implicit (backward Euler) scheme and Newton iterations, with an adaptive time step of 43s to 36 min. For speed the model is compiled into C++ by the OHQ code generator, which reproduces the interpreter’s solution; one year of simulation takes a few minutes on one processor core.

5 Data

All inputs are public. Table 1 lists the sources; the maps below show how they enter the model.

Table 1: Data sources.

Model input	Source
Terrain, drainage network	USGS 3DEP 10 m DEM and 1 m lidar; NHDPlus High Resolution
Land cover, imperviousness	Chesapeake Conservancy 1 m land use/land cover (2024 edition)
Connected impervious area	Sutherland (2000) relation, checked against public storm-drain inventories
Soils	NRCS SSURGO; van Genuchten parameters from Rosetta3
Channels	regional Piedmont bank-full relations; 1 m lidar; USGS field measurements and continuous record
Reservoirs	National Inventory of Dams; 1 m lidar
Combined-sewer area	DC GIS, combined sewer system area
Rainfall (calibration, validation)	NOAA Analysis of Record for Calibration (AORC), hourly, 1 km
Rainfall (operational)	NOAA MRMS multi-sensor QPE (hourly, 1 km); Open-Meteo forecast
Reference ET	gridMET (daily, 4 km); Open-Meteo FAO-56 reference ET (recent days and forecast)
Discharge and stage	USGS National Water Information System

5.1 Land cover and imperviousness

Land-cover fractions of each sub-catchment come from the 1 m Chesapeake land-cover data (Figs. 12 and 13). Total impervious area is converted to directly connected impervious area with the “average” Sutherland relation $DCIA = 0.1 TIA^{1.5}$ (in percent; Fig. 14), which was checked independently against 52 000 storm-drain inlets from the public inventories of Montgomery County, the Maryland State Highway Administration, the municipalities and DC Water (Figs. 15 and 16); the calibration then scales it watershed-wide.

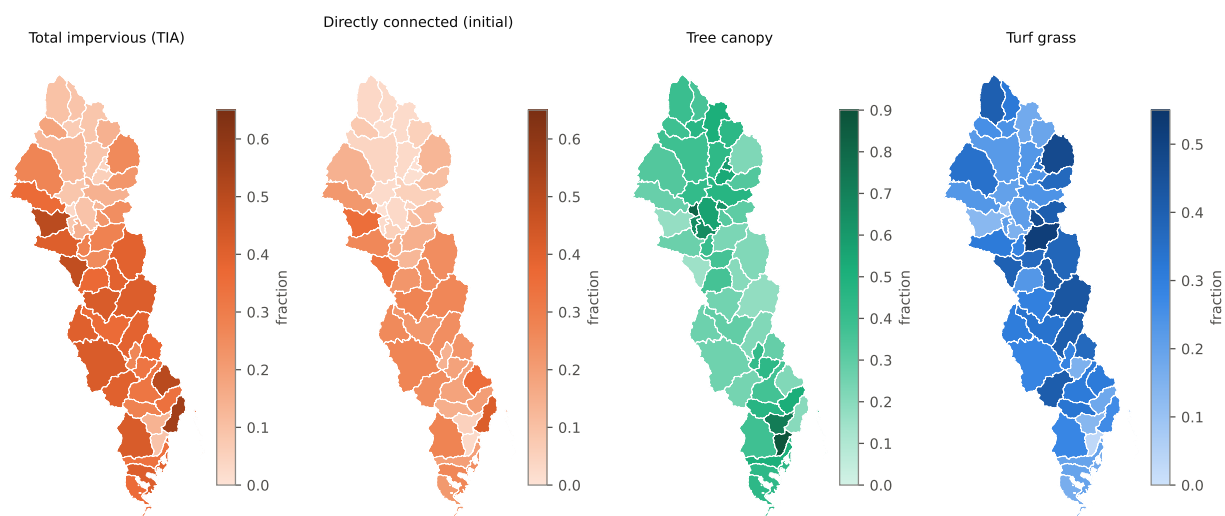


Figure 12: Land-cover fractions by sub-catchment.

1 m land cover, 3 × 3 km around Rock Creek at Sherrill Dr

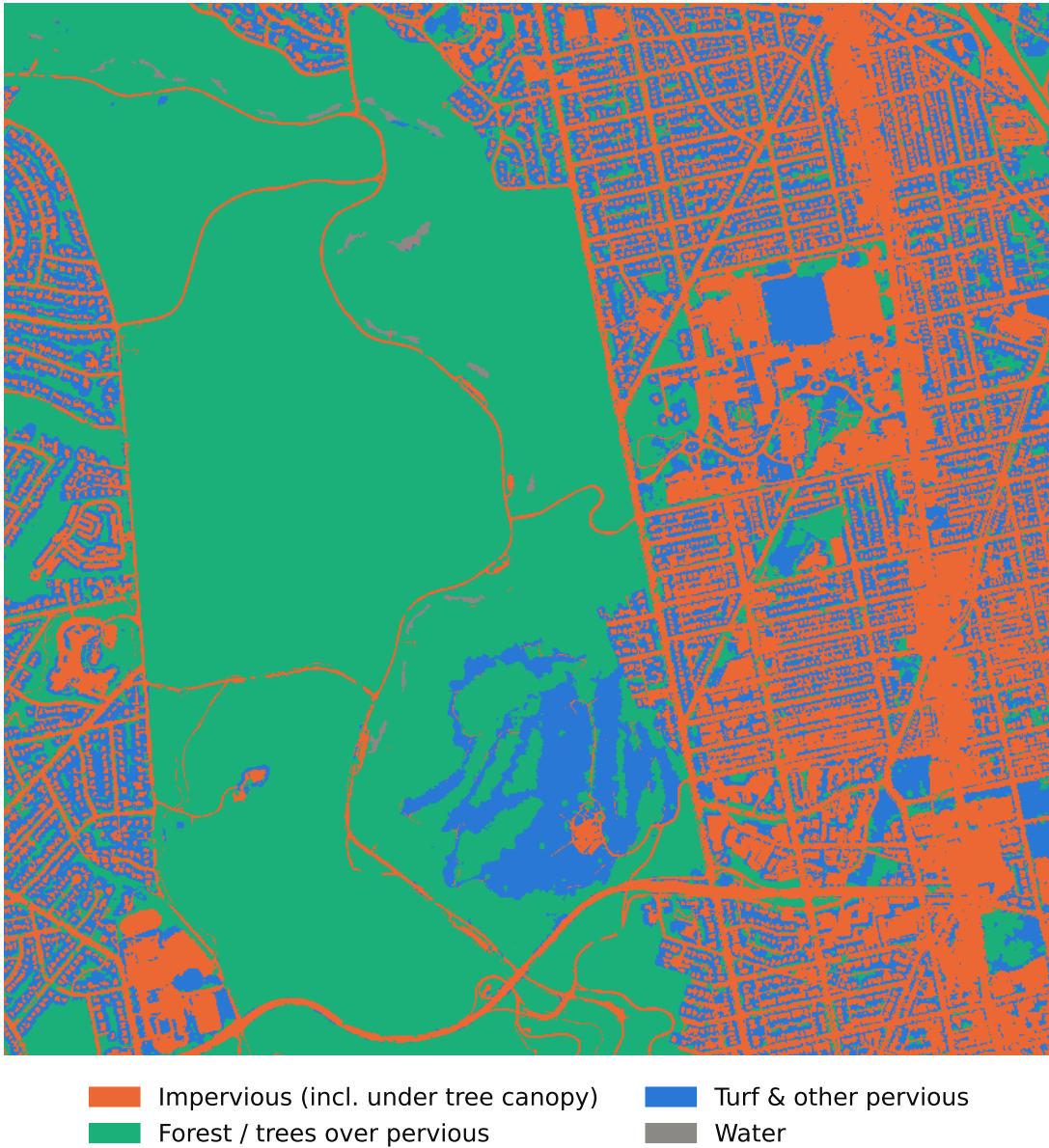


Figure 13: 1 m land cover near the Sherrill Drive gage, grouped into three classes.

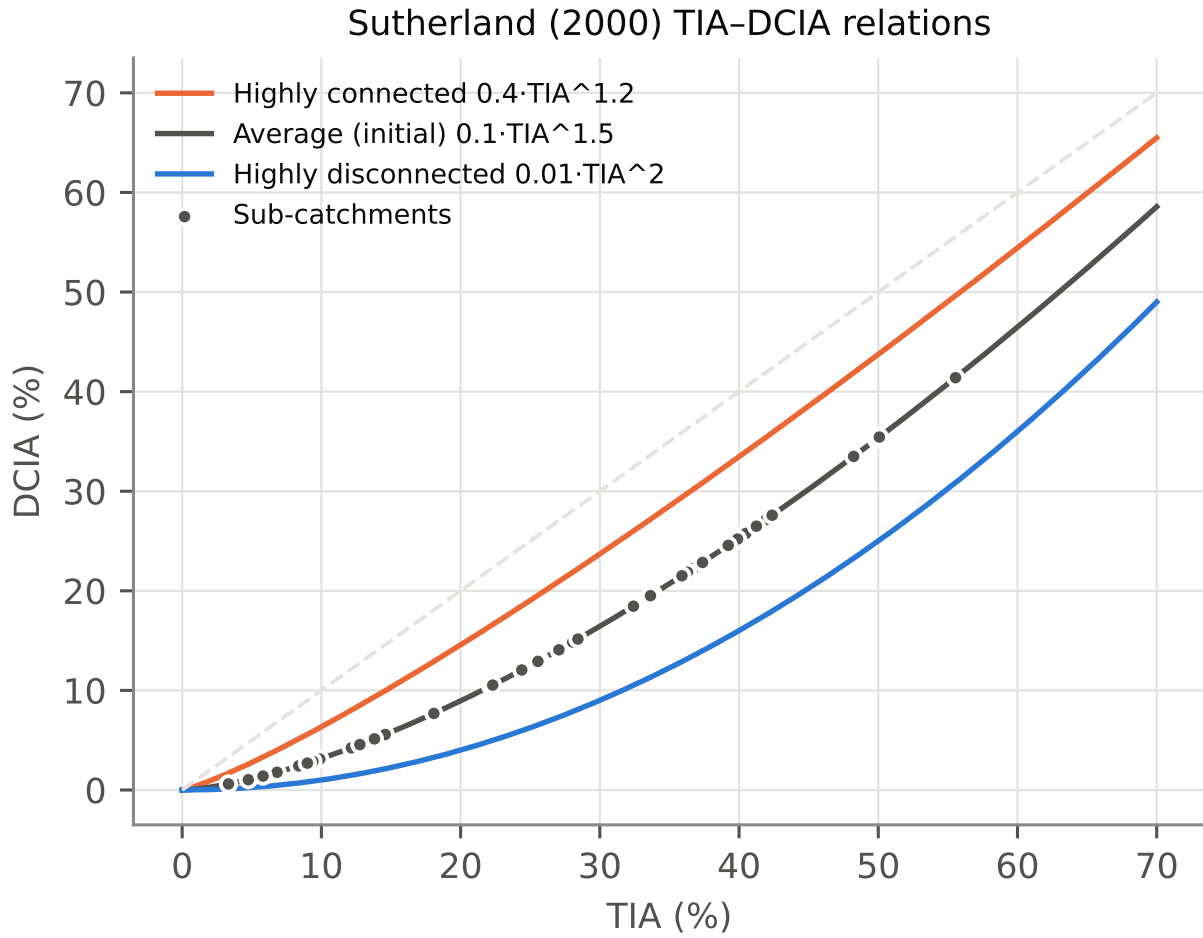


Figure 14: Total and directly connected impervious area: the Sutherland (2000) relations and the sub-catchment values.

Storm-drain inventories (DC: DC Water catch basins only)

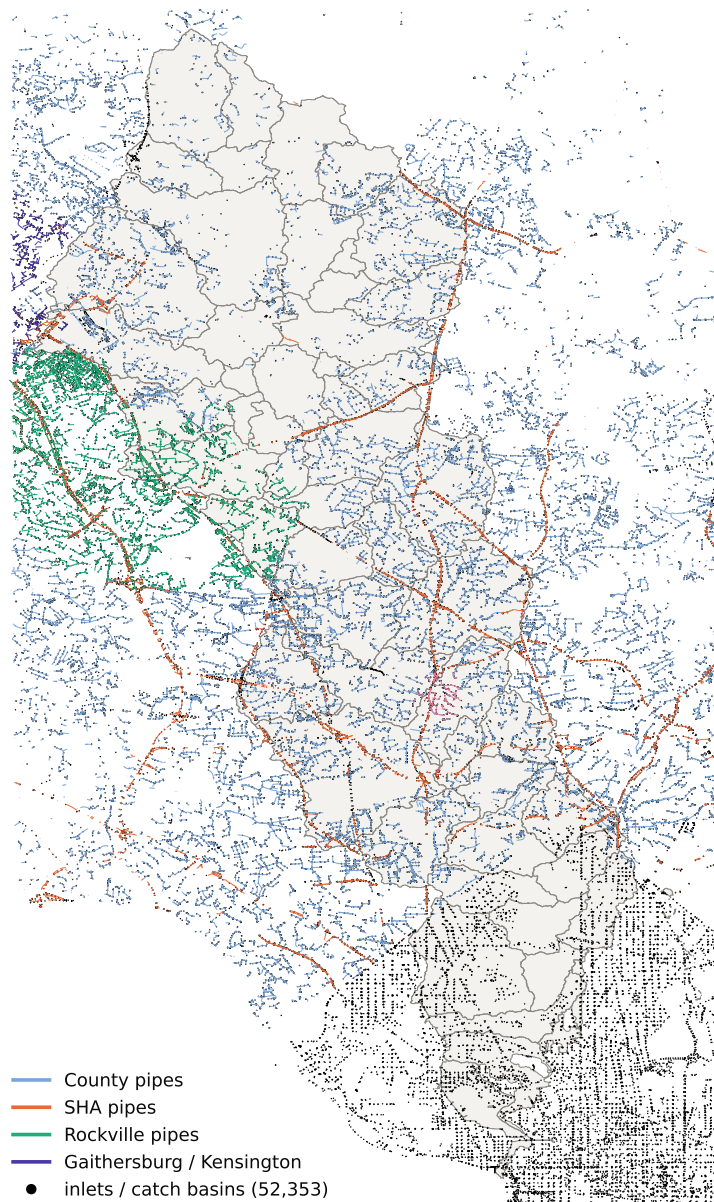


Figure 15: Public storm-drain inventories used to check the connected impervious area.

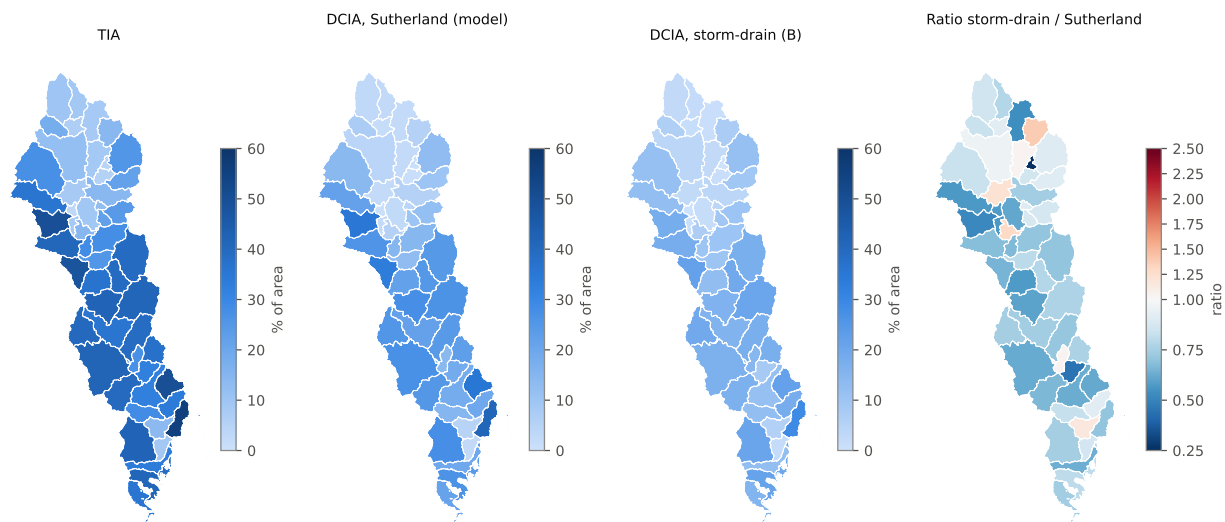


Figure 16: Connected impervious area from the storm-drain inventories, by sub-catchment.

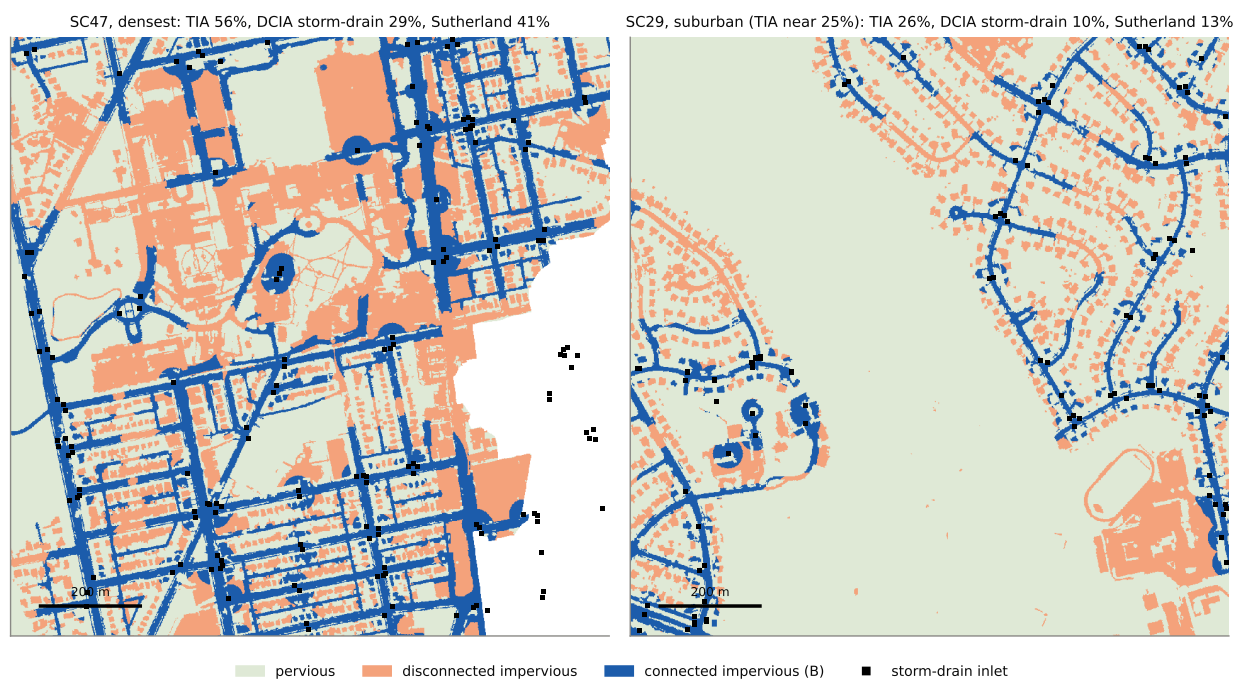


Figure 17: Connected and disconnected impervious surfaces at 1 m with the storm-drain inlets, in a dense and a suburban sub-catchment.

Combined-sewer areas removed from the contributing area

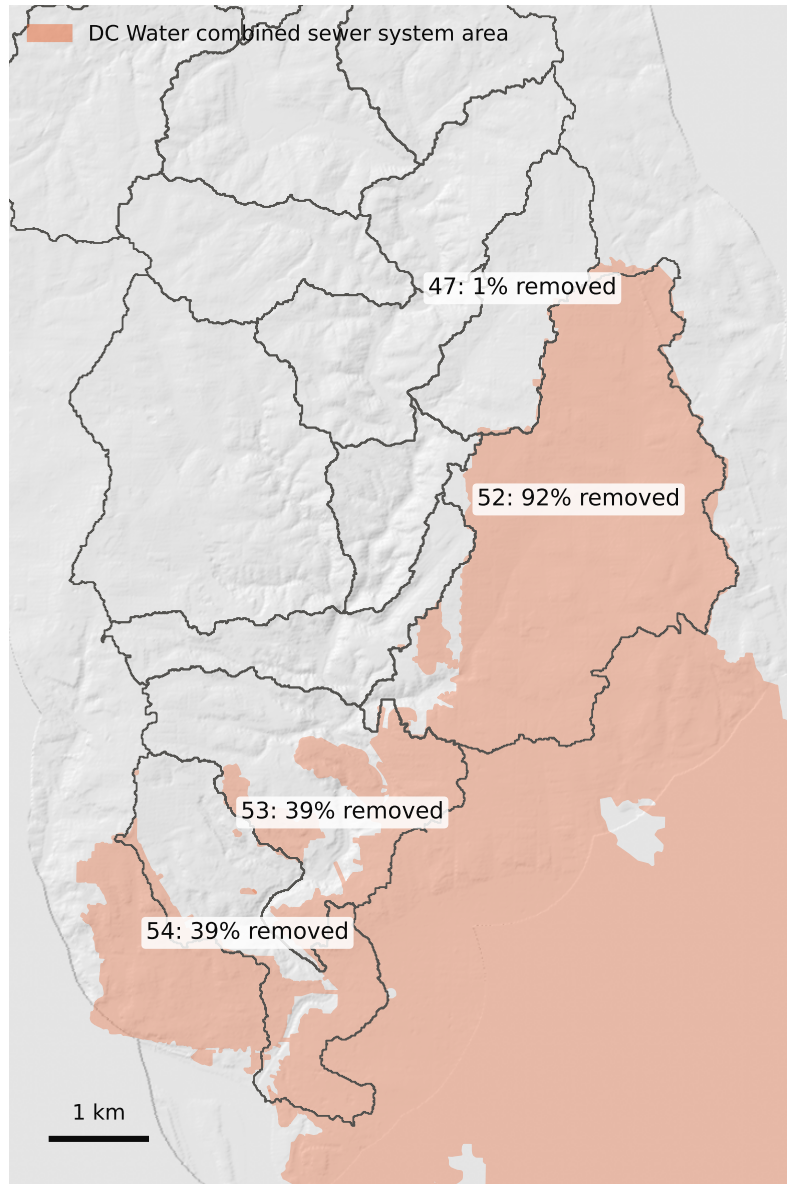


Figure 18: DC Water combined-sewer area, excluded from the contributing area.

5.2 Soils and terrain

Soil hydraulic properties of the three layers are area- and depth-weighted SSURGO values, with van Genuchten parameters from the Rosetta3 pedotransfer functions (Fig. 19). Overland-flow length, width and slope and the internal reach lengths come from the terrain (Fig. 20).

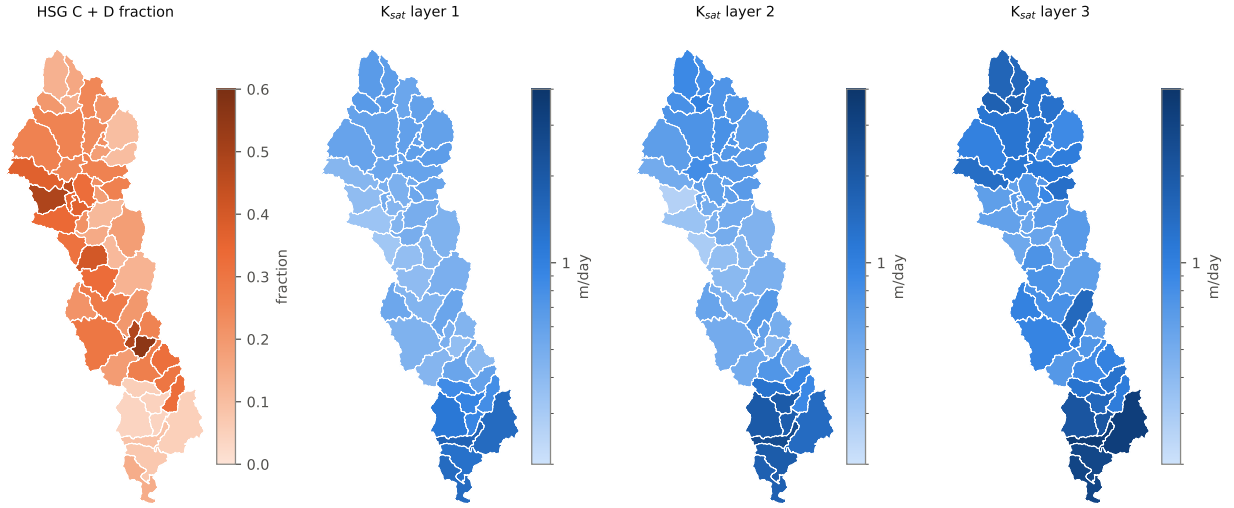


Figure 19: Hydrologic soil group C+D fraction and saturated conductivity of the three soil layers.

Terrain attributes by sub-catchment (10 m DEM)

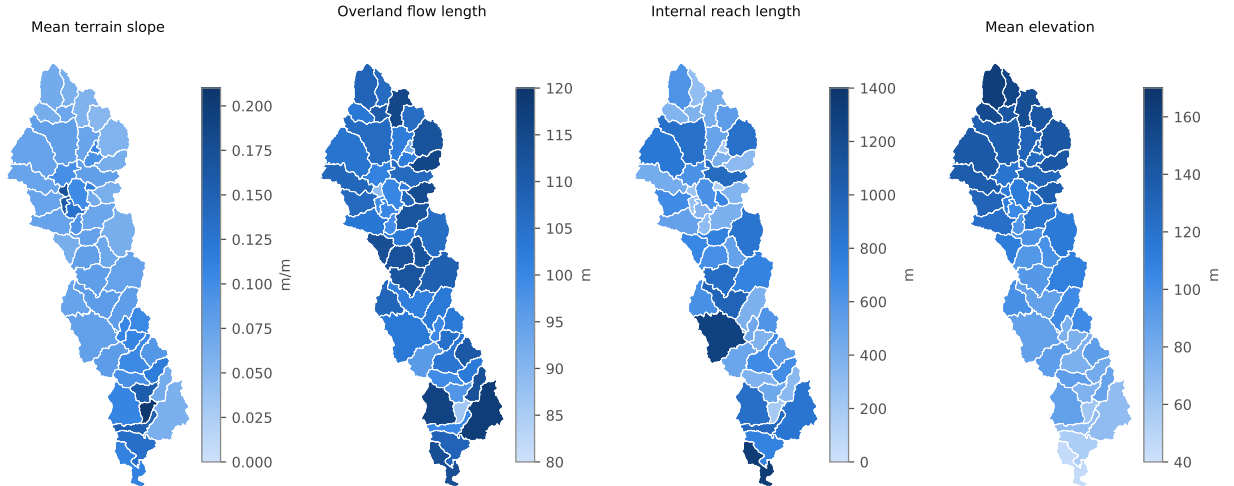


Figure 20: Terrain attributes by sub-catchment.

5.3 Meteorological forcing

Rainfall and reference evapotranspiration are averaged over each sub-catchment with the overlap areas of the grid cells (Fig. 22). The model was calibrated and validated with AORC rainfall; operationally it uses MRMS. Over the year in which both are available (October 2024 to September 2025) the two agree closely in timing (correlation 0.98 for hourly and 0.99 for daily basin rainfall), but AORC is 8.4% wetter in total (Fig. 21); most of the difference is light rain below 0.5 mm/h,

and for storm rainfall the difference is 2–8 % with no trend with storm size. MRMS is therefore multiplied by 1.08 to stay consistent with the calibration. gridMET and the Open-Meteo reference ET agree to within about 1 % over their overlap.

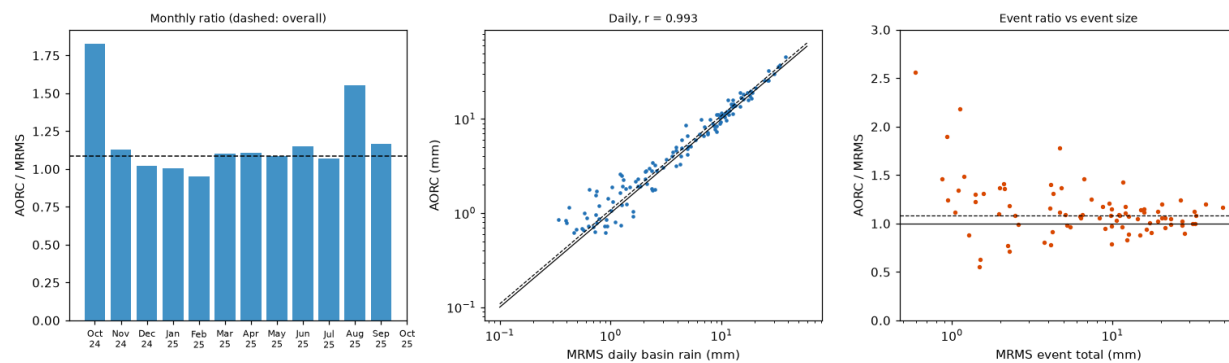


Figure 21: AORC (calibration rainfall) versus MRMS (operational rainfall), basin mean, October 2024 to September 2025: monthly ratio, daily totals and the ratio for individual storms versus storm size. Dashed lines: the overall ratio, 1.084.

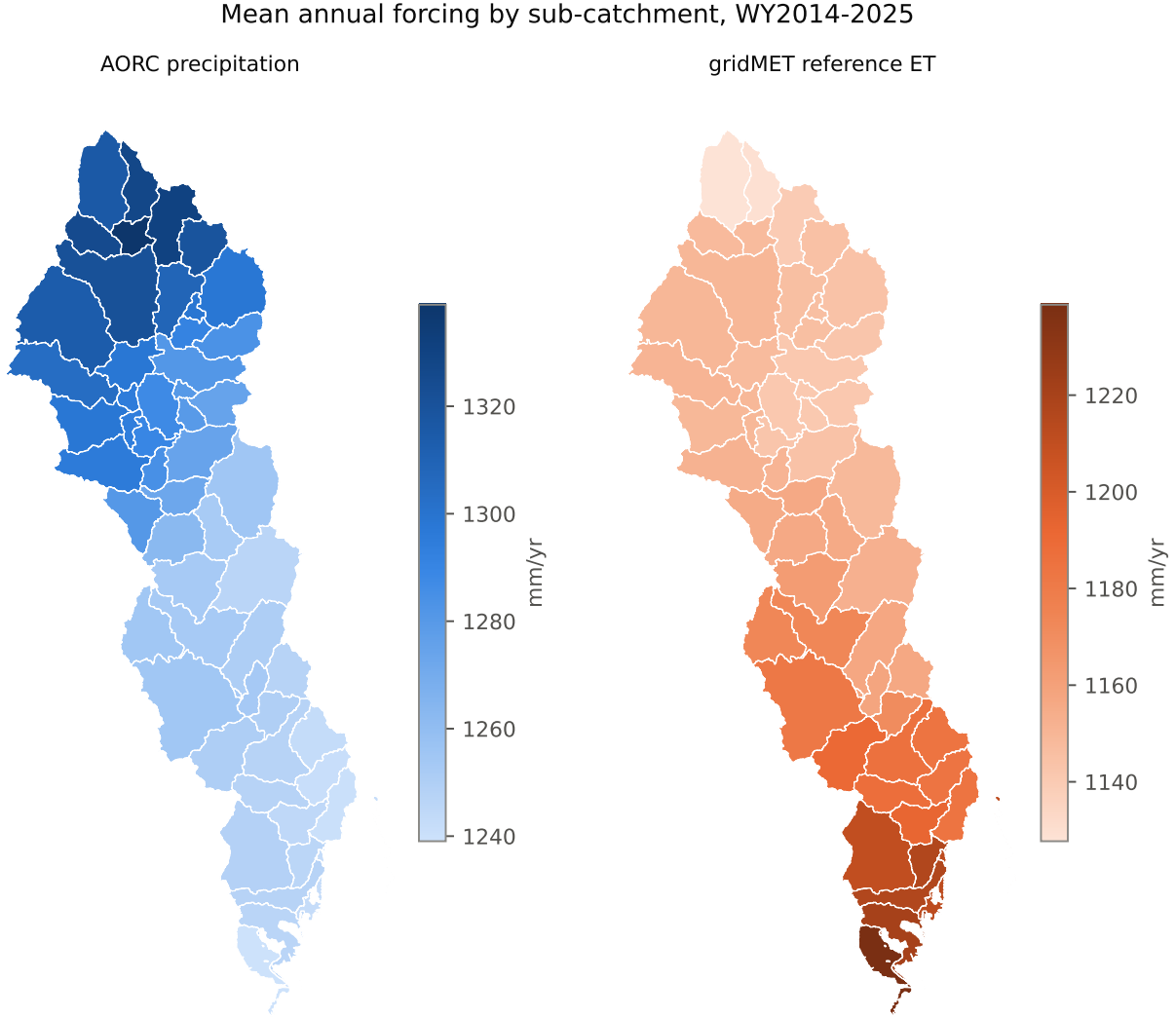


Figure 22: Mean annual AORC precipitation and gridMET reference ET by sub-catchment, WY2014–2025.

6 Calibration

Twelve parameters were estimated, each applied identically to all 54 sub-catchments (Table 2), so that the spatial differences between sub-catchments come entirely from the data above. They were fitted to discharge and stage at Rock Creek at Sherrill Drive (USGS 01648000) and at Joyce Road (01648010) over water years 2018–2019 (hourly values during storms, daily values otherwise), after a 14-month spin-up. The objective is the likelihood of the log-transformed observations; a genetic algorithm searched the parameter space and Levenberg–Marquardt refined the result. Turkey Branch (01647850) was kept out of the calibration as an independent test.

Table 2: Calibrated parameters (final values).

Parameter	Role	Value
Soil conductivity multiplier	scales SSURGO K_s of all layers	4.42
Surface conductivity multiplier	additional factor on the top layer (compaction)	0.019
Connected-impervious multiplier	scales the Sutherland DCIA	1.22
Crop coefficient, top layer	K_c	0.67
Crop coefficient, deeper layers	K_{c2}	0.29
Regional aquifer conductivity	m/d	0.023
Regional aquifer specific yield	–	0.060
Near-stream fraction	f_r	0.054
Near-stream specific yield	–	0.39
Near-stream and bank conductivity	m/d	1.02
Channel roughness	Manning n	0.050
Reservoir riser rating multiplier	–	0.84

7 Validation

The calibrated model was run without any change from October 2016 to September 2025 and compared with the USGS record of water years 2020–2025, which was not used in calibration, at the two main-stem gages and at Turkey Branch (Table 3 and Figs. 23 and 24). On the main stem the model reproduces flow with an NSE of 0.73–0.78 and stage with 0.80–0.82 over the six years, with no water year below 0.60 (Table 4), and peak flows within 3–9 % on average. Total volume is 5–16 % high, almost entirely from dry-weather flow: the flow-duration curves agree above 20 % exceedance and the model is about twice the observed flow in the lowest 20 %. At Turkey Branch, which was never used in calibration, flow is reproduced with an NSE of 0.72 and its flow-duration curve is matched over the whole range; its stage is reproduced poorly (NSE 0.21): between storms the stage of this small channel varies over only a few centimetres, and the model’s low-flow stage is 3–5 cm high (its low flows are too high, and the measured low-flow stage has fallen by about 3 cm over the six years), which is enough to dominate the measure.

Table 3: Validation, WY2020–2025 (model run continuously from October 2016 with the calibrated parameters).

Gage		NSE	logNSE	Volume ratio	Peak ratio
Rock Creek at Sherrill Dr (01648000)	flow	0.73	0.74	1.16	1.03
	stage	0.82	0.82		
Rock Creek at Joyce Rd (01648010)	flow	0.78	0.77	1.05	0.91
	stage	0.80	0.77		
Turkey Branch (01647850), not calibrated	flow	0.72	0.46	1.15	1.00
	stage	0.21	0.23		

Table 4: Flow NSE by water year, validation period.

Gage	2020	2021	2022	2023	2024	2025
Sherrill Dr	0.60	0.78	0.76	0.74	0.82	0.71
Joyce Rd	0.71	0.80	0.79	0.77	0.80	0.81
Turkey Branch	0.75	0.68	0.74	0.64	0.79	0.69

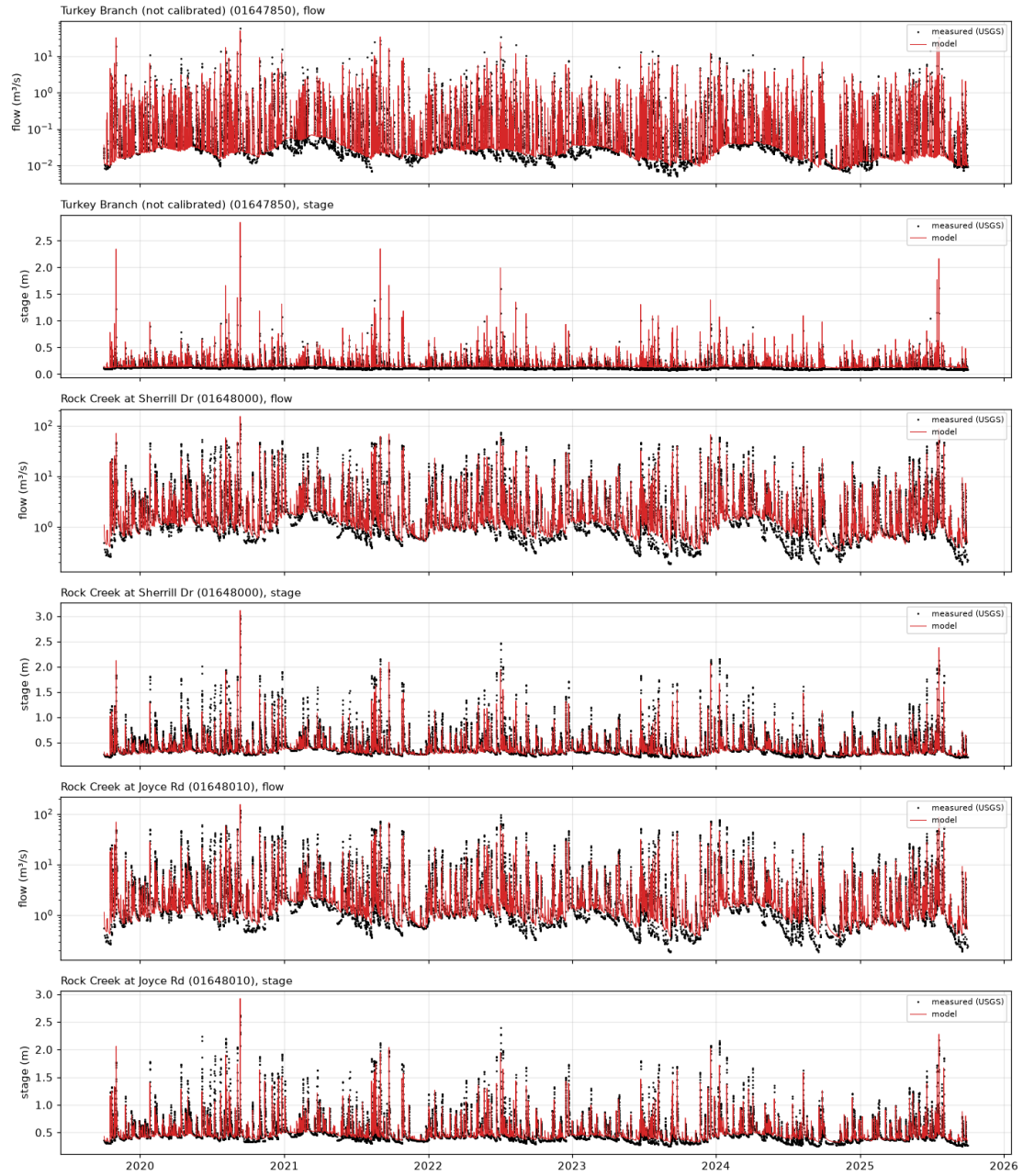


Figure 23: Validation, WY2020–2025: measured (USGS, black) and modelled (red) flow (log scale) and stage.

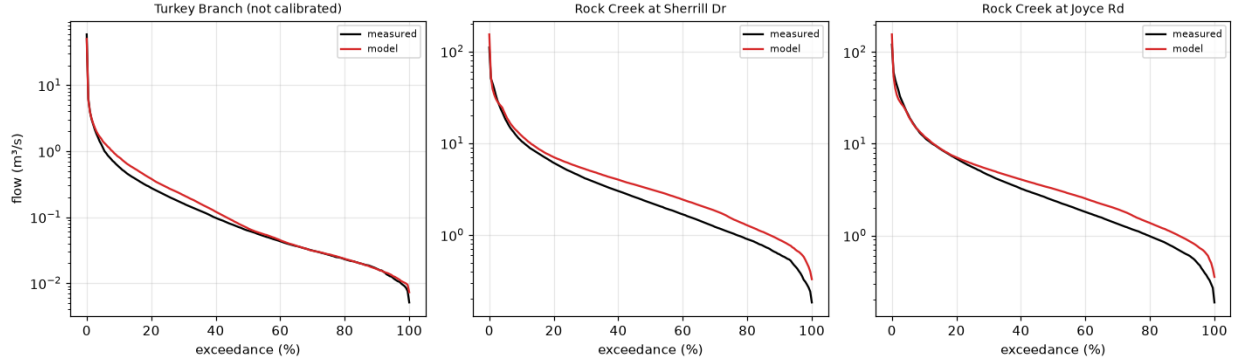


Figure 24: Flow-duration curves, WY2020–2025.

Fit measures. NSE: Nash–Sutcliffe efficiency of the flows (1 is perfect, 0 is no better than the mean); logNSE: the same on the logarithms, which weights low flows and recessions; volume ratio: total model over observed flow; peak ratio: model over observed mean of the highest 20 % of flows; KGE: Kling–Gupta efficiency, which combines correlation, variability and bias (1 is perfect).

7.1 Comparison with the National Water Model

NOAA’s National Water Model (NWM) simulates flow on about 2.7 million stream reaches across the United States, Rock Creek included (Cosgrove et al., 2024). It is built on the WRF-Hydro system (Gochis et al., 2020): the Noah-MP land surface model on a 1 km grid, terrain routing, a conceptual groundwater “bucket” per catchment that supplies baseflow, and Muskingum–Cunge channel routing, with parameters calibrated on a national set of basins. Its retrospective simulation (version 3.0, 1979 to January 2023) is driven by the same AORC rainfall as our calibration and validation runs, which makes it a like-for-like benchmark. Table 5 compares the two models at the times of the USGS observations over WY2020–January 2023, years not used to calibrate our model.

Both models reproduce the storm peaks (Fig. 25). The difference is in the rest of the hydrograph: the National Water Model keeps dry-weather flow about three times higher than observed on the main stem and fourteen times higher at Turkey Branch, where its flow-duration curve levels off near 0.2–0.3 m³/s (Fig. 26). This model reduces that bias to about 1.8 times on the main stem and reproduces Turkey Branch, which it was not calibrated on, with a KGE of 0.81 against 0.11. It scores higher than the National Water Model on every measure at every gage, in this period and in WY2018–19 as well. The comparison is with the National Water Model’s retrospective simulation, which does not use gage observations; its operational forecasts adjust the simulated flow to real-time gage data at the start of each forecast, so forecast skill by lead time is the next comparison. The National Water Model is designed for flood forecasting across the country; the comparison shows what a watershed-specific model built from local data adds for small urban streams and for low flows.

Table 5: Our model and the NOAA National Water Model v3.0 retrospective (both AORC-forced) against USGS, WY2020–January 2023 (not used to calibrate our model). Low/High 20 %: model over observed mean of the lowest/highest 20 % of observed flows. NWM has one reach (22338431) for Sherrill Dr and Joyce Rd.

Gage	Model	NSE	logNSE	KGE	Volume	Low 20 %	High 20 %
Rock Creek at Sherrill Dr	this model	0.71	0.76	0.80	1.13	1.8	1.02
	NWM v3.0	0.60	0.55	0.72	1.21	3.0	0.98
Rock Creek at Joyce Rd	this model	0.76	0.79	0.85	1.01	1.8	0.90
	NWM v3.0	0.66	0.60	0.77	1.07	3.0	0.85
Turkey Branch	this model	0.73	0.46	0.81	1.13	4.6	0.98
	NWM v3.0	0.49	−0.60	0.11	1.86	14.1	1.24

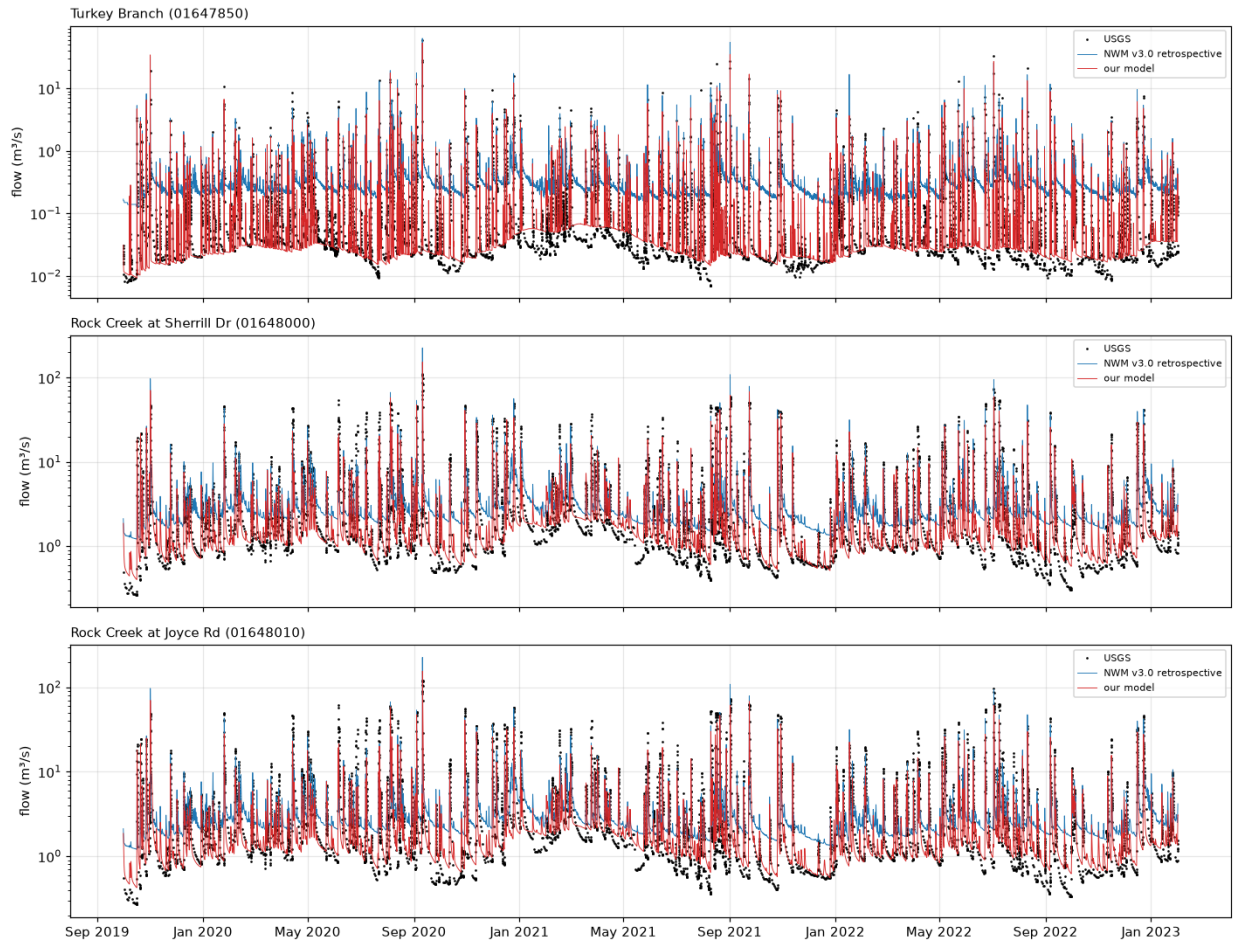


Figure 25: Flow at the three gages, WY2020–January 2023: USGS (black), the National Water Model v3.0 retrospective (blue) and this model (red), log scale.

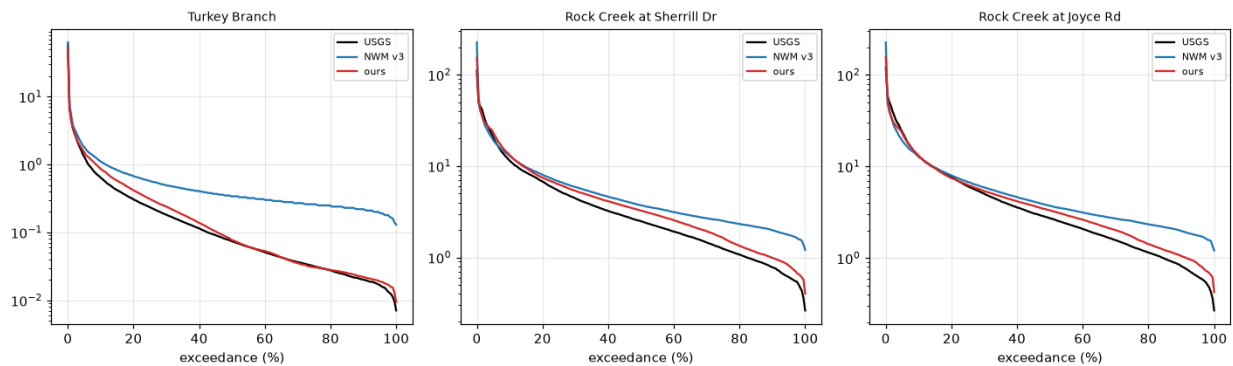


Figure 26: Flow-duration curves, WY2020–January 2023: USGS, the National Water Model v3.0 retrospective and this model.

8 Limitations

- **Low flows.** In dry weather the model keeps main-stem flow higher than observed (about 1.5–2 times in the lowest 20 % of flows); forecasts of drought flows should be read with this in mind.
- **Small tributaries.** Parameters are watershed-wide. Turkey Branch, more impervious and with a sparser channel network than the watershed as a whole, is reproduced less well than the main stem.
- **Forecast rainfall** is the largest source of forecast error, especially for convective summer storms; the twin currently runs a single forecast without ensemble spread.
- **Not represented:** combined-sewer overflows, snow accumulation and melt, withdrawals and transfers, and the operation of the reservoir gates beyond their fixed spillways.
- **No data assimilation** yet: the model state is not corrected with the real-time observations.

9 Software and reproducibility

The model is built and run with open-source software: OpenHydroQual (<https://github.com/ArashMassoudieh/OpenHydroQual>; Massoudieh et al., 2023; Massoudieh and Shakouri, 2025), the digital-twin engine OpenHydroTwin (<https://github.com/ArashMassoudieh/OHTwin>) and the watershed twin and viewer OpenWatershedTwin (<https://github.com/ArashMassoudieh/OpenWatershedTwin>). Every input file of the model is produced by numbered scripts from the public data listed in Table 1.

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